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Nanoindentation derived stress–strain properties of dental materials

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ABSTRACT

Objectives. The aim of the study is to investigate the stress–strain response of different dental materials, especially dental brittle materials, and compare them with enamel.

Methods. A nano-based indentation system (Ultra Micro-Indentation System, UMIS-2000, CSIRO, Australia) was used to determine the indentation stress–strain response of two kinds of dental ceramics (Cerec®2 Mark II and Vita VM9), one kind of dental alloy (Wiron® 99) and healthy enamel. A spherical indenter was used to test the materials with nanometer and micro-Newton displacement and force resolution. Assuming the elastic modulus remained constant, a plot of contact pressure versus contact strain, H–a/R, of each material was obtained.

Results. By comparing the H–a/R curve of the different materials with enamel, it can be concluded that only the metallic alloy, has similar stress–strain response as enamel. Dental ceramics showed much higher yield stress response than enamel. VM9, a porcelain veneer component of crown/bridge structure, is slightly softer than its core, Mark II. The yield point for Mark II and VM9 are nearly 10 and 7 GPa, respectively, and approximately 2 GPa for Wiron alloy and enamel.

Significance. H–a/R curves provide a new method to compare the mechanical properties of different dental materials. From the standpoint of structural reliability, strong and tough materials with primarily elastic response, such as toughened ceramics are required to enable dental crown/bridges to have long term reliability. On the other hand, materials with too high hardness or yield response may damage opposing teeth during occlusal contact. Future studies may establish a relationship between stress and strain property and abrasive wear of dental material.

1. Introduction

General mechanical property characterization of materials has become an essential requirement for their successful application as structural elements. For metallic materials, these properties are well described and numerous handbooks are available enabling comparison for structural design purposes. In the field of dental materials, such properties, especially stress–strain behavior, are not available for most materials. This deficiency is considered to be a major short coming especially with the wide spread use of ceramics in dentistry, the elastic–plastic mechanical properties of these materials is of considerable interest.

Among all the mechanical properties, stress–strain behavior is one of the most important because this property determines the deformation response of the material. At the same

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time, the ductility and yield ability are also important for the application of the material as a structural element undergoing contact loading. Moreover, this property has a close relationship with the wear or abrasive nature of the material, which is very important for dental material. For the new generation of dental ceramics which exhibit quasi-plastic behavior, quantifying this property becomes more important.

The instrumented indentation technique has become a popular method to measure the mechanical properties of different materials. However, most investigations use a sharp indenter such as Berkovich indenter from which the hardness and elastic modulus may be determined. Few investigators have chosen a spherical tipped indenter. In the dental materials area, a number of studies have been published on the measurement of dental materials and tooth properties using nanoindentation [1–6], but no other investigation with spherical indenter tip was found apart from a study by Poolthong et al. [7] to investigate the properties of enamel and dentine. A sharp tipped indenter does not allow the transition from elastic to plastic behavior of a material to be deduced because these indenters produce a nominally constant plastic strain impression. A spherical tip, in contrast, with increasing depth of penetration, develops increasing contact stress so that the elastic to plastic transition response and the determination of the contact stress–strain property of a material may be investigated [8,9].

The use of spherical tipped indenters has a long history for characterizing metallic materials [10]. Tabor found a simple empirical relationship between contact pressure or hardness and contact strain (contact radius/indenter radius, a/R). This approach and the relationships between hardness and true stress as well contact strain and true strain has been an area of further study such as that by Matthews [11] and only recently has been analytically confirmed by Hill et al. [12]. Recently, Lawn and colleagues have investigated the behavior of dental ceramics with a spherical indenter [13–16]. The latter research has mainly concentrated on the macro-scale damage mode of brittle material and multilayer structures.

With the development of high resolution instrumented indentation devices, new methods are available to measure the mechanical properties of materials. Oliver and Pharr [17] proposed a method based on the contact mechanics analysis developed by Sneddon [18] to analyze any axi-symmetric indentation generated P–h curves to determine the Hardness and Elastic Modulus mechanical properties of materials. Field and Swain [19] developed an alternate method to analyze force-displacement data using a spherical tipped indenter. Later, Oliver and Pharr [20] and Suganuma and Swain [21] verified the equivalence of the two approaches.

In this paper, a new method was developed to measure the indentation stress–strain relationship of dental materials with small spherical indenter tips and various dental materials as well as enamel were investigated.

2. Theory

The models chosen to interpret the indentation data were that of Hertz [10], Oliver and Pharr [17] and Tabor [22]. Unfortunately, on a micro-scale level, it is impossible to produce

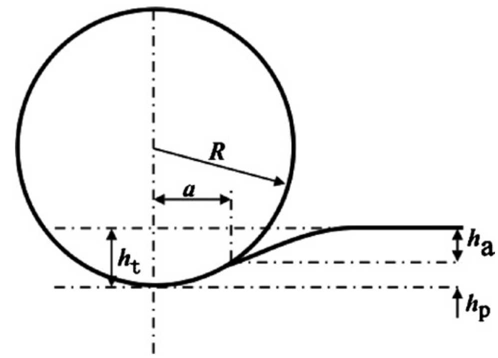

Fig. 1 – Schematic illustration of contact between a spherical indenter and a flat specimen.

a perfect spherical indenter with a constant radius. In the case of a spherical tipped indenter Hertz's analysis provides an elastic contact relationship between indenter and samples (see Fig. 1).

The radius of contact circle, a, can be calculated as,

$$a^3 = \frac{3}{4}\frac{PR}{E_{\mathrm{eff}}}, \tag{1}$$

in which $1/E_{\mathrm{eff}} = (1-\nu^2)/E + (1-\nu'^2)/E'$, E, E' and ν, ν' are Young's modulus and Poisson's ratio of specimen and indenter (usually diamond), P is the contact load and a is the contact radius.

The total depth of penetration, h_t, is given by:

$$h_t^3 = \left(\frac{3}{4E_{\mathrm{eff}}}\right)^2 \frac{P^2}{R}. \tag{2}$$

Contact pressure, P_m, can be calculated by the definition of the pressure,

$$P_m = \frac{P}{\pi a^2}. \tag{3}$$

Hertz also established the simple relationship between the elastic total displacement into the sample, h_t, and the contact depth, h_p,

$$h_t = 2h_p. \tag{4}$$

Combining Eqs. (1) and (3), the relationship between the contact pressure, P_m, and the contact radius, a, is,

$$P_m = \left(\frac{4E_{\mathrm{eff}}}{3\pi}\right)\frac{a}{R}, \tag{5}$$

where E_{eff} is the reduced or indentation modulus and R is the nominal radius of the indenter tip. This relationship establishes the linear relationship between contact pressure and contact strain, a/R, during the elastic loading of a material.In the case of elastic–plastic response of materials to pointed indenters Oliver and Pharr [17] recommended a simple method to fit the nanoindentation P–h unloading curve, namely;

$$P = \alpha(h - h_r)^m, \tag{6}$$

where P is the indenter load, h the elastic displacement, h_r the residual impression depth of the indenter, while α and m are constants. This method applies to all axi-symmetric indenters, including spheres. With this equation, the elastic contact stiffness, dP/dh, can be calculated from the upper portion of the unloading data.

There is a constant relationship between the elastic contact stiffness, dP/dh, the projected area of contact, A, and the reduced or indentation modulus E_{eff},

$$\frac{dP}{dh} = \beta \frac{2}{\sqrt{\pi}} E_{eff} \sqrt{A} \tag{7}$$

In which $A = \pi a^2$, β is a constant dependent upon indenter shape. For a spherical indenter, $\beta = 1$.

In this context, by assuming that the E modulus of a material is known, the radius of contact area, a, at certain loading force, can be calculated from Eq. (7), namely

$$a = \frac{dP/dh}{2E_{eff}}. \tag{8}$$

At the same time, a relationship between the indenter radius, R, contact radius, a, and contact depth h_p, can be derived from Fig. 1,

$$a = \sqrt{2h_p R - h_p^2} \tag{9}$$

By combining Eqs. (8) and (9), it is not difficult to show,

$$\frac{a}{R} = \frac{((dP/dh)/E_{eff})h_p}{((dP/dh)/2E_{eff})^2 + h_p^2} \tag{10}$$

Based on the definition of hardness, H,

$$H = \frac{P_{max}}{A} \tag{11}$$

The nanoindentation system can record the P–h response of a material from which P_{max}, h_p and dP/dh maybe automatically calculate. Thus, a relationship between H and a/R curve of the sample may be determined.

In the case of a conical tipped indenter Tabor [22] showed that

$$\varepsilon_r \approx 0.2 \tan\beta \tag{12}$$

$$\sigma \approx \frac{P_m}{3} \tag{13}$$

are appropriate expressions for representative strain and stress where β is the angle between the indenter flank and the original surface. Matthews [11] on the other hand suggested that the divisor in Eq. (13) should be 2.8 rather than 3.

In spherical indentation, strain increases continuously with indentation impression depth, and an appropriate equivalent expression is

$$\varepsilon_r \approx \frac{0.2a}{R} \tag{14}$$

where $\tan\beta \approx \sin\beta \approx a/R$.

In this study, the approach of Tabor, namely the H–a/R curve generated for a material shall be regarded as the contact stress–strain curve.

3. Experimental methods

Before the test, a standard fused silica specimen was chosen to calibrate the Berkovich and spherical indenters. More than 100 tests were done with both the Berkovich and the spherical indenter within the load range of the system to calibrate the indenters. For the Berkovich indenter, the software generated the calibrated area function automatically. For the spherical indenter, the contact area A was calculated and the contact radius a determined versus contact depth of penetration, h_p, which represents the shape of the tip. The spherical diamond tipped indenter [Synton, Switzerland] had a nominal indenter radius of 10 μm.

Two dental ceramics, one dental cast alloy and healthy enamel of a permanent tooth were selected to illustrate the approach. The dental ceramics were Cerec®2 VitaBLOCS® Mark II CAD/CAM ceramic (A3, 5C, V5-12) and its veneer porcelain Vita VM9 (Base Dentine, 1M1) [VITA Zahnfabrik, Germany]. The dental alloy was Wiron® 99 [BEGO, Germany] with a composition in weight percentage as follows: Ni 65%, Cr 22.5%, Mo 9.5%, Nb, Si, Fe, Ce each less than 2%. The tooth was an orthodontically extracted healthy permanent premolar.

The Mark II and Wiron blocks were cut into pieces with 1.5 mm thickness using a low speed saw [Isomet, Buehler Ltd., Lake Bluff, USA] under tap water irrigation. The VM9 porcelain specimen was prepared following the manufacturer's instruction into a block with the thickness of approximate 2 mm. The healthy premolar tooth was embedded with a cold-curing epoxy resin [Epofix, Struers, Copenhagen, Denmark] and the lingual surface of the tooth was chosen to do the test. All the specimens were polished with two machines [Grinder-Polisher, Buechler UK Ltd., Coventry, England; RotoPol-22, Struers, Copenhagen, Denmark] to 1 μm diamond polishing paste. Once the tooth had been prepared, it was stored in distilled water at room temperature. Previous investigations by Poolthong [23] using spherical tipped indenters over the same load range as investigated in this study, established the long term stability of the hardness and elastic modulus of enamel stored at 4 °C in de-ionized distilled water. All the other samples were stored at room temperature in dry laboratory conditions.

The indentation experiments were performed using a nano-based indentation system [Ultra Micro-Indentation System, UMIS-2000, CSIRO, Australia]. The finished specimens were mounted on metal bases with wax using a paralleling machine [Leitz, Wetzlar, Germany]. The mounting base contained a strong magnet to ensure adequate contact was obtained with the test base in the UMIS. All the tests of ceramics and alloys were done under dry condition while enamel was tested in distilled water.

Elastic modulus and hardness of specimens were measured with the calibrated Berkovich indenter. The force applied during testing ranged from 200 to 50 mN with a decrement of 10 mN for each sample. Following completion of each test array, the UMIS software calculated values for

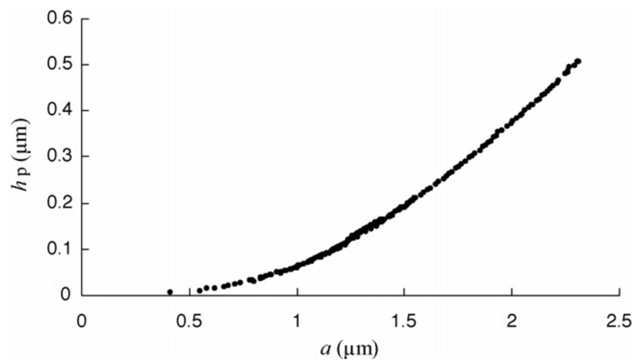

Fig. 2 – h_p–a curve of spherical tip indenting fused silica.

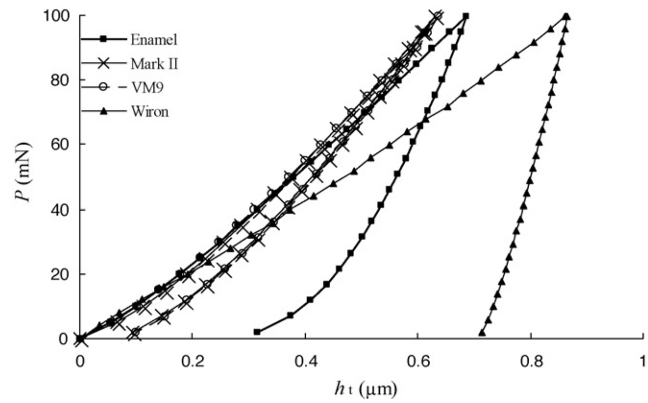

Fig. 3 – Comparison of P–h_t curve of three different dental materials and enamel under 100 mN loading force.

elastic modulus and hardness as a function of penetration depth, h_p, for each indentation.

All the samples were indented with the spherical indenter at loading forces from 1 to 400 mN. The distance between two indents on the surface of the sample was more than 40 μm to avoid the influence of residual stresses from other impressions. The H–a/R curves were generated using the methods outlined above.

After nanoindentation, an SEM (XL-30, Philips, The Netherlands) was used to check the damage of indents. All the specimens were ultrasonic cleaned with ethanol and then gold coated. Both secondary electron detector and back scattered electron detectors were selected for the observations.

4. Results

4.1. *Shape of spherical indenter*

From the fused silica test data, h_p–a curve were generated for the spherical indenter used in this study (see Fig. 2). This figure illustrates the effective shape of the spherical indenter. The indenter used in this work has a parabolic shape with a changing radius from 10 μm to nearly 6 μm with increasing contact depth, h_p.

4.2. *Elastic modulus and hardness of dental materials*

Table 1 summarizes the results of elastic modulus and hardness for each sample. The alloy has the highest elastic modulus among all the samples. Enamel has a higher elastic modulus than ceramics. The veneer part of the ceramic structure, VM9 has a slightly lower elastic modulus than its core, Mark II. The ceramic materials have the highest hardness values of nearly 10 and 7 GPa, which are well above that of the enamel. The metallic alloy has the lowest hardness of only 4.1 GPa.

Using Tabor's approach and the relationship $\tan\beta \approx \sin\beta \approx a/R$, we compared the effective hardness for the spherical indenter with that of the Berkovich indenter having an effective cone angle of 70.3°. The values obtained were similar for the hardness acquired from Berkovich indenter and spherical indenter as shown in Table 1.

4.3. *Sample spherical load–displacement curves*

Load–displacement data is the basic response acquired from the depth-sensing indentation system and is the loading–unloading history of the sample. Fig. 3 illustrates typical load–unloading curves of the four materials indented to a maximum force of 100 mN. Dental ceramics showed mostly elastic recovery upon unloading while the metallic alloy exhibited almost a totally plastic response. Enamel, on the other hand, displayed a response between the elastic behavior of the ceramics and plastic behavior of the metal investigated.

4.4. *SEM observation of spherical indentations on the surface of ceramics*

Figs. 4 and 5 are SEM images of spherical indenter residual impressions on the surface of Mark II and VM9 specimen. The loading force for both indents was 450 mN. Both pictures show cracks generated by the spherical indenter. For Mark II ceramic, the indenter produced not only cone cracks but also radial cracks. VM9, in contrast, had a damage mode of mainly cone cracks. From Fig. 5 we can also see that Mark II ceramic contains many particulates in its structure. It is interesting to notice that indentation induced radial cracks pass through these particles. VM9, on the other hand, has a homogeneous structure.

Table 1 – Nanoindentation properties of different dental materials

Materials	Enamel	Mark II	VM9	Wiron
Elastic modulus (GPa)	105.5 ± 3	78.9 ± 2.94	65.52 ± 2.89	199.54 ± 12.5
Nanoindentation hardness (GPa)	5.58 ± 0.35	10.64 ± 0.46	9.5 ± 0.35	4.1 ± 0.17
Effective hardness (GPa)	5.3	11.5	8.5	4.9

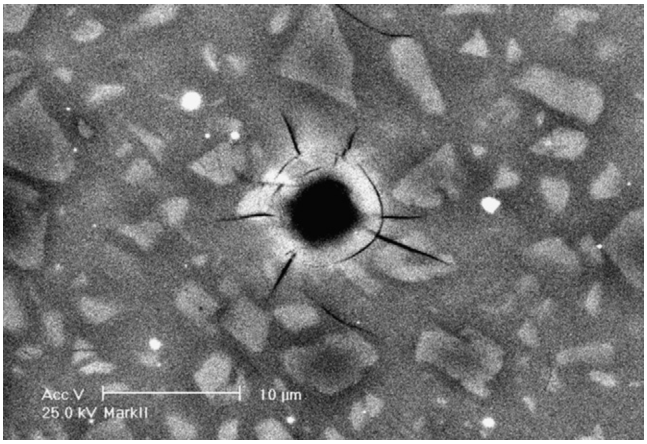

Fig. 4 – Back scattered SEM image of surface indent on Mark II sample using the spherical tipped indenter.

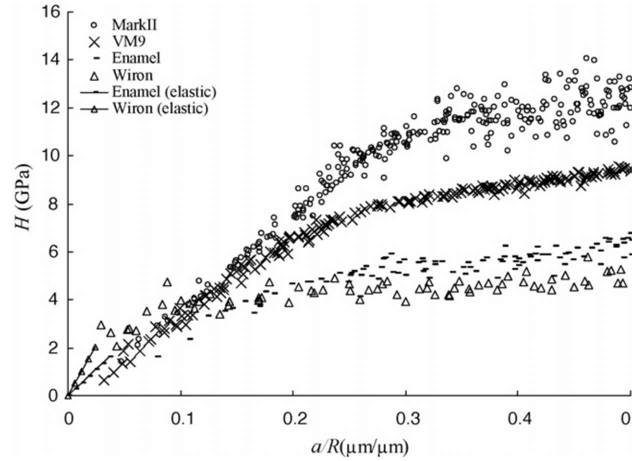

Fig. 6 – Nanoindentation stress–strain curve of different dental materials.

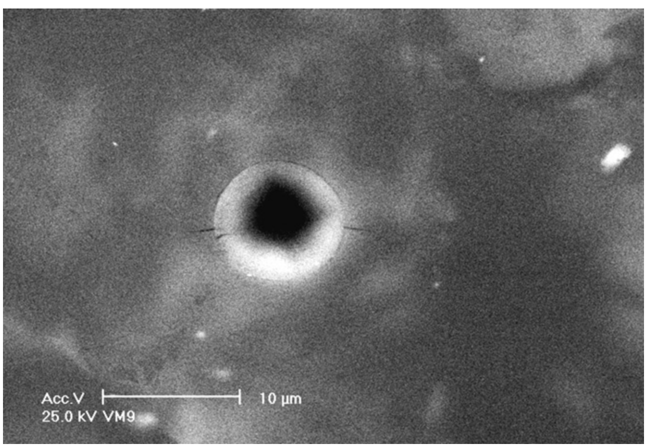

Fig. 5 – Back scattered SEM image of surface indent on VM9 sample using the spherical tipped indenter.

4.5. *Contact stress–strain response of dental materials*

Fig. 6 shows the contact stress–strain response of the different dental materials as well as enamel. In these curves each point is the result of one complete load–unload indentation test. From the data points, only a limited elastic response range was found for enamel and alloy Wiron® 99. Therefore, we deduced elastic response of these two materials from Hertz equation (Eq. (2)). They are represented by lines named "enamel (elastic)" and "Wiron (elastic)" in the figure. Although enamel and metallic alloy have higher elastic modulus, dental ceramics had much higher "yield" points in their contact stress–strain responses. Based on these contact stress–strain curves, Mark II, a core ceramic structure, has a higher yield stress than VM9. At the same time, the dental alloy shares similar stress–strain response as enamel.

4.6. *Contact yield point of dental ceramics*

From Eq. (4), within the elastic limit, the contact depth h_p and total depth h_t have a relationship of $h_p/h_t = 0.5$. When the stress–strain response of the material enters the plastic range, the aforementioned relationship no longer holds and the ratio of h_p over h_t will increase. Therefore, directly, from the figure of h_p/h_t–H, we can read the yield strength of the material (see Fig. 7). Within the low contact pressure or elastic region, both ceramics show a trend of decreasing h_p/h_t values from nearly 0.5 to 0.45. Nevertheless, in an effort to make use of the acquired data, we chose to consider the point at which the shape of curve changed dramatically to represent the yield point from elastic to elastic–plastic behavior. The arrow in the figure indicates these points. The yield contact pressure is approximately 7 GPa for VM9 porcelain and near 10 GPa for Mark II ceramic.

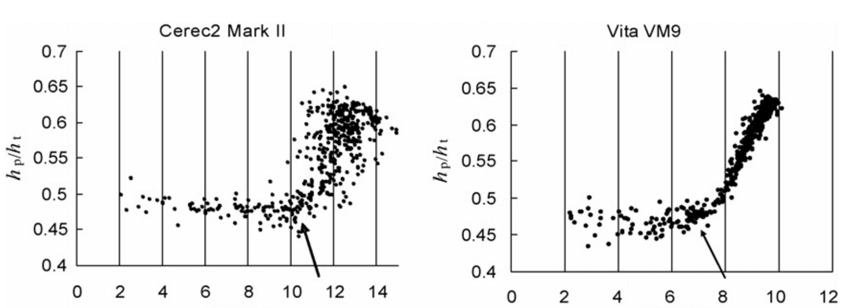

Fig. 7 – Plots of h_p/h_t–H curves of dental ceramics investigated.

5. Discussion

5.1. *Indenter tip*

The stresses and deflections arising from the contact between two elastic solids are of particular interest to those undertaking indentation testing. From Fig. 1, it is evident that the contact depth, h_p, versus contact radius, a, may be used to define the nominal shape of the indenter. Fig. 3 shows the effective shape of the indenter we used in this study. It is found that the indenter has a parabolic shape rather than spherical. That is, the radius of spherical indenter is changing with contact depth, h_p. However, Oliver and Pharr [20] have shown that the method used to calculate the contact pressure is applicable to any axi-symmetric indenter, including this parabolic one.

Based on Tabor's work, the stress–strain function $H(a/R)$ is a unique function for a given deformable material, regardless of the nature of deformation and independent of sphere size [22]. However, reports [24–26] have shown that in the micro and nano scale indentation field, indentation size effect (ISE) is a major influence on test results. That is to say, different size of indenter may lead to different results. In the case of crystalline materials this phenomenon can be explained by the concept of "geometrically necessary dislocations". Based on this concept, with the changing of the tip radius, the onset of the plastic response of the stress–strain curve may change. Moreover, enamel and dentine, which are typical anisotropic materials with multiple characteristic scale lengths, may exhibit ISE. These may become an important and interesting area for future research.

5.2. *Elastic modulus and hardness*

The results of the elastic modulus and hardness of enamel are a little higher than the value from Habelitz et al. [27] where an elastic modulus of 87.5 ± 2.2 GPa and the hardness of 3.9 ± 0.3 GPa were reported. This difference may come from the different depth of enamel below the surface investigated. Cuy et al. [28] have shown that the mechanical response of enamel depends upon location, chemical composition, and prism orientation. The hardness value of enamel decreases from more than 6 GPa at the top surface of the tooth to less than 3 GPa near the Enamel-Dentine Junction area. The current investigation was done very near the top lingual surface of the premolar tooth. Based on the conclusion of Cuy et al., the hardness value of this area should be somewhat higher than the inner part of the enamel. On the other hand, since the functional area of tooth and denture should be the occlusal surface area, the investigation of mechanical property of this area will have more practical significance than inner parts.

For the dental alloy Wiron® 99, the elastic modulus is very similar to that listed in the manufacture's instruction, 205 GPa. In Lawn's paper [14], the elastic modulus of Mark II is 68 GPa. David and LoPresti [29] also reported that the elastic modulus of this material are also much closer to those of tooth structure. All these results coincide with the outcomes of the present study. This suggests that the UMIS system and the area function we created are reliable.

5.3. *Load–displacement curve*

The use of spherical tipped indenters enables the initial elastic response prior to commencement of elastic–plastic displacement during loading. On the other hand, the unloading procedure is generally regarded as a purely elastic process. From the energy point of view, the area between the loading and unloading curves represents the irrecoverable energy loss (W_i) because of plastic or inelastic deformation. The total energy (W_t) is the area under the loading curve, which represents the total energy expended in forming the indentation, and includes both the elastic and plastic components. The energy absorption ratio is calculated from $(W_i/W_t) \times 100\%$. Wiron alloy shows the onset of energy absorption at low stresses and has the highest energy absorption ratio while the ceramic materials have the lowest ratio. This means that the alloy has the capacity to absorb severe localized loading by deformation thereby reducing the contact stresses with an opposing tooth during contact. In contrast, the ceramic materials have limited ability to deform and exhibit minimal energy loss until yielding occurs at very high-localized stresses which are more likely to be harmful to the opposing teeth.

5.4. *SEM results*

Cerec®2 Mark II ceramic is a feldspathic ceramic with a composition of SiO_2, Al_2O_3, K_2O, Na_2O. SiO_2 and Al_2O_3 sintered together resulting in a microstructure with particulates of 1–7 μm diameter, as observed by SEM. Fig. 4 shows that these particles are well distributed in the glass matrix. No microcracks were found along the interface of the particle and the matrix. It is these particles which appear to cause the scatter of the indentation data in Fig. 6. VM9, in contrast, has a more homogenous structure than Mark II as shown in Fig. 5 and exhibits less data scatter.

The damage mode observed in Figs. 4 and 5 are similar to Lawn's results [13–15], which showed that cone cracks are a common damage mode and in less brittle, tougher ceramics, the 'quasi-plasticity' of the materials may lead to a 'yield' zone of distributed shear microcracks.

5.5. *Stress–strain response*

In the theoretical section above, we mentioned Tabor's equation to convert H–a/R values into σ–ε curves. Should we follow Tabor's idea and regard the later value as the true stress–strain curve? To answer this question, we should make clear two things. Firstly, Tabor's expression is an empirical relationship based on experimental observation for metallic materials. It is still not clear if we can apply this relationship to brittle materials. Secondly, the contact stress distribution is complex and it is difficult to compare this type of stress–strain relationship with uni-axial tensile/compression tests. Herbert et al. [30] for aluminum alloy has shown that the indentation derived stress–strain relationship has a big difference to the uni-axial one. This maybe anticipated as a consequence of the indentation size effect and the respective volumes of material involved in the two types of tests. Recently, some researchers have made progress in relating spherical indentation stress–strain curve with the uni-axial tests for metallic materials [31–33]. It is uncertain whether this method is universal or not and if it can be extended into the ceramic field. Therefore, in this work, we retain the data as H–a/R curves and these plots represent the *indentation* stress–strain response of different materials.

In Table 1, we compared the hardness values from Berkovich and spherical indenters by using Tabor's equation. The spherical indentation results were generated at the contact strain in Eq. (14). The results are similar for the two kinds of indenters and the small difference may come from the approximation of the relationship that is $\tan\beta \approx \sin\beta \approx a/R$.

From the standpoint of structural reliability and integrity, two issues are important. First, we need materials with higher strength as well as better elastic response. Material with high elastic to plastic transition point (yield point) can bear a greater force within its elastic limits. In oral functional environment, because ceramic has higher yield point than enamel, we can predict that the ceramic structure (crown/bridge) can sustain higher bite force than enamel provided it does not fracture. Therefore, this kind of material and its structure is strong enough to serve as a denture. Moreover, Mark II, as the core material, has higher yield strength and elastic limit than its veneer part, VM9. This coincides with the structural design criteria, which requires a stronger core to support the whole structure. The higher yield strength of Mark II appears to come from the ceramic particles dispersed in the glass matrix. Of course, other factors that influence the integrity of a structure include fatigue and fracture toughness. Second, when it comes to the contact damage and surface fatigue wear, the condition may be totally different. At the micro-scale level, the actual contact surface consists of numerous asperities because of surface roughness. For each contact pair within the contact area, the two opposite parts bear the same force. If the difference of mechanical properties, especially the stress–strain property, between the contact pair is too big, the soft one will be destroyed while the hard one remains intact. Therefore, the component with lower stress–strain response will wear. Clinically, if Mark II serves as an inlay of posterior teeth, it will be in contact directly with opposite tooth. During the surface contact period, the contact pressure on the contact point may exceed the elastic limit of enamel while still remaining within the elastic range of ceramic. Under this condition, permanent plastic deformation or even damage may occur on the surface of enamel. Thus, the surface of the opposite tooth will be at risk of excessive wear. However, other researches [34,35] have suggested that mechanical properties are only one factor that influence the wear between enamel and dental material. Moreover, it is becoming increasingly clear that hardness may not be a good predictor of potential abrasive of dental materials [36]. Wear properties of material, especially of ceramics, also depend upon microstructure, fracture toughness, and the mechanism of contact damage [36–38]. In general, contact damage characteristics are known to be a function of the fracture toughness of ceramics, micro-structural scale (grains, filler phases, pores), and local property variations in its microstructure [36,38]. The stress–strain response provides a novel means to reevaluate the study of wear and contact damage. Future work shall attempt to correlate the stress–strain property with the abrasive ability of dental materials.

5.6. *Yield point*

The results in Fig. 7 indicate that the ratio of the contact depth to total indenter displacement is a function of the contact pressure rather than a constant within the elastic range of the ceramic materials. Unfortunately, the ratio is slightly less than anticipated from Hertz contact theory. This is similar to what Herbert et al. saw with aluminum alloy 6061-T6 [30]. Understanding the cause of these different results will be an integral part of future work. However, the trend of the curves also provides some indications about the yield of the materials investigated. The arrows illustrate the inflexion point of the ratio of the contact to total indenter displacement, which may be regarded as the yield point of the materials. The yield contact pressure is approximately 7 GPa for VM9 porcelain and near 10 GPa for Mark II ceramic, which coincides with the trend of the H–a/R curves.

For the Wiron alloy and enamel, plastic or inelastic deformation dominates their indentation response, as even with an initial loading force of 5 mN, the contact pressure exceeds the elastic limit of these materials. The initial ratio of h_p/h_t is larger than 0.6 and we cannot obtain the elastic part of the H–a/R curve with this spherical indenter. With a larger spherical tip, it should be possible to investigate the elastic response range of these materials. However, with the help of Hertz equation, we can predict the transition point of these plastic materials as drawn in Fig. 6. The transition stress from elastic to plastic response of these materials occurs at approximately 2 GPa.

6. Conclusion

This study used a small spherical tipped indenter to determine the indentation stress–strain properties of different dental materials as well as enamel. The spherical indentation contact pressure results at comparable indentation strain are comparable with Berkovich indentation results. Mark II and VM9 have contact yield pressures of nearly 10 and 7 GPa, respectively; whereas the metallic alloy Wiron® 99 and enamel had more comparable stress–strain response with less well defined contact yield pressures of nearly 2 GPa. This approach provides a unique means for determining the localized mechanical response of dental materials and further study will investigate whether a relationship exists between indentation stress–strain behavior and abrasive ability.

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